Nonlinear Finite Element Methods

Lecture 2: Newton Iteration and the Tangent System

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Lecture outline

1. Total potential energy, its variations, and the residual.
2. Newton's method for a scalar equation.
3. Linearization of the hyperelastic variational residual.
4. Finite element residual, tangent matrix, and correction boundary data.
5. A stretch example and checks on the nonlinear solve.

The aim is to obtain the linear system solved at each Newton iteration, including its entries and its right-hand side.

Recall: the hyperelastic residual

On the fixed reference body, $\boldsymbol{F}(\boldsymbol{u}) = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}$ and $\boldsymbol{P} = \partial \Psi / \partial \boldsymbol{F}$. With prescribed reference loads independent of displacement,

$$a_{\mathrm{hyp}}(\boldsymbol{u}; \boldsymbol{v}) = \int_{\Omega_0} \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u})) : \operatorname{Grad} \boldsymbol{v} \, \mathrm{d}V_0,$$

$$\ell_{\mathrm{hyp}}(\boldsymbol{v}) = \int_{\Omega_0} \boldsymbol{b}_0 \cdot \boldsymbol{v} \, \mathrm{d}V_0 + \int_{\Gamma_{0t}} \overline{\boldsymbol{t}}_0 \cdot \boldsymbol{v} \, \mathrm{d}A.$$

Our starting equation is

$$\boxed{R(\boldsymbol{u}; \boldsymbol{v}) = \ell_{\mathrm{hyp}}(\boldsymbol{v}) - a_{\mathrm{hyp}}(\boldsymbol{u}; \boldsymbol{v}) = 0 \quad \forall \boldsymbol{v} \in \mathcal{V}_0.}$$

Here $\boldsymbol{u} = \overline{\boldsymbol{u}}$ on Γ_{0u}, while every $\boldsymbol{v} \in \mathcal{V}_0$ is zero there. We seek an admissible deformation with $\det \boldsymbol{F} > 0$. A trial displacement generally has a nonzero residual.

Total potential energy and the variational problem

With prescribed reference loads independent of displacement, define

$$\Pi_{\mathrm{hyp}}(\boldsymbol{u}) = \int_{\Omega_0} \Psi(\boldsymbol{F}(\boldsymbol{u}))\,\mathrm{d}V_0 - \ell_{\mathrm{hyp}}(\boldsymbol{u}).$$

For $\boldsymbol{v} \in \mathcal{V}_0$, use $\boldsymbol{F}(\boldsymbol{u}+\epsilon\boldsymbol{v}) = \boldsymbol{F}(\boldsymbol{u}) + \epsilon\,\mathrm{Grad}\,\boldsymbol{v}$ and $\boldsymbol{P} = \partial\Psi/\partial\boldsymbol{F}$. The first variation is

$$\begin{aligned}
\delta\Pi_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}) &= \frac{\mathrm{d}}{\mathrm{d}\epsilon}\Pi_{\mathrm{hyp}}(\boldsymbol{u}+\epsilon\boldsymbol{v})\bigg|_{\epsilon=0} \\
&= \int_{\Omega_0} \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u})) : \mathrm{Grad}\,\boldsymbol{v}\,\mathrm{d}V_0 - \ell_{\mathrm{hyp}}(\boldsymbol{v}) \\
&= a_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}) - \ell_{\mathrm{hyp}}(\boldsymbol{v}) = -R(\boldsymbol{u};\boldsymbol{v}).
\end{aligned}$$

Thus stationarity is exactly the original variational problem:

$$\boxed{\delta\Pi_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}) = 0\ \forall \boldsymbol{v} \in \mathcal{V}_0 \quad\Longleftrightarrow\quad a_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}) = \ell_{\mathrm{hyp}}(\boldsymbol{v})\ \forall \boldsymbol{v} \in \mathcal{V}_0.}$$

A stationary point need not be a minimum.

Second variation and the residual derivative

For twice-differentiable Π_{hyp} and fixed $\boldsymbol{v}, \boldsymbol{w} \in \mathcal{V}_0$, define

$$\delta^2 \Pi_{\mathrm{hyp}}(\boldsymbol{u}; \boldsymbol{v}, \boldsymbol{w}) := \left. \frac{\mathrm{d}}{\mathrm{d}\epsilon} \delta \Pi_{\mathrm{hyp}}(\boldsymbol{u} + \epsilon \boldsymbol{w}; \boldsymbol{v}) \right|_{\epsilon=0}.$$

Since $R(\boldsymbol{u}; \boldsymbol{v}) = -\delta \Pi_{\mathrm{hyp}}(\boldsymbol{u}; \boldsymbol{v})$,

$$\begin{aligned} \delta_u R(\boldsymbol{u}; \boldsymbol{v})[\boldsymbol{w}] &= \left. \frac{\mathrm{d}}{\mathrm{d}\epsilon} R(\boldsymbol{u} + \epsilon \boldsymbol{w}; \boldsymbol{v}) \right|_{\epsilon=0} \\ &= - \left. \frac{\mathrm{d}}{\mathrm{d}\epsilon} \delta \Pi_{\mathrm{hyp}}(\boldsymbol{u} + \epsilon \boldsymbol{w}; \boldsymbol{v}) \right|_{\epsilon=0} \\ &= -\delta^2 \Pi_{\mathrm{hyp}}(\boldsymbol{u}; \boldsymbol{v}, \boldsymbol{w}). \end{aligned}$$

In UFL, `u` is the current displacement, `v` the test function, and `w` the trial direction:

```
residual = -ufl.derivative(Pi, u, v)
tangent = ufl.derivative(residual, u, w)
```

Newton's method: one scalar equation

Consider $r(z) = z^2 - 2 = 0$. Let z^k be an approximation and Δz^k its correction. Linearize about z^k:

$$r(z^k + \Delta z^k) \approx r(z^k) + r'(z^k)\Delta z^k.$$

Set this linear approximation to zero:

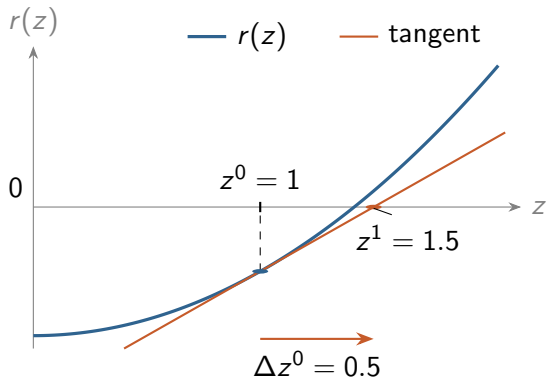
$$\boxed{r'(z^k)\Delta z^k = -r(z^k), \qquad z^{k+1} = z^k + \Delta z^k.}$$

Starting from $z^0 = 1$,

$$r(1) = -1, \qquad r'(1) = 2, \qquad 2\Delta z^0 = 1, \qquad z^1 = 1.5.$$

Each iteration solves a linear correction equation and updates the approximation.

A tangent equation predicts the next approximation



The tangent at $z^0 = 1$ is

$$r(1) + r'(1)(z - 1) = 2z - 3.$$

Its zero gives $z^1 = 1.5$.
Repeating the update gives

$$1.5, \quad 1.416667, \quad 1.414216.$$

The tangent must be evaluated again at each new iterate.

Convergence of scalar Newton iteration

For $r(z) = z^2 - 2$, the update is

$$z^{k+1} = z^k - \frac{(z^k)^2 - 2}{2z^k} = \frac{1}{2}\left(z^k + \frac{2}{z^k}\right).$$

Starting at $z^0 = 1$ gives:

k	z^k	$\lvert r(z^k)\rvert$
0	1.000000	1.0000
1	1.500000	0.2500
2	1.416667	6.9444×10^{-3}
3	1.414216	6.0073×10^{-6}

These iterates converge to $\sqrt{2}$. Starting at $z^0 = 0$ would fail because $r'(0) = 0$. Newton's method needs a suitable initial iterate and a solvable tangent equation.

Newton correction for the displacement field

Linearize $R(\boldsymbol{u}; \boldsymbol{v}) = 0$ with the test function held fixed.

The same correction idea for displacement

For hyperelasticity, seek $R(\boldsymbol{u}; \boldsymbol{v}) = 0$ for every $\boldsymbol{v} \in \mathcal{V}_0$. Hold $\boldsymbol{v}$ fixed and define the variation with respect to $\boldsymbol{u}$ in a direction $\boldsymbol{w} \in \mathcal{V}_0$:

$$\delta_u R(\boldsymbol{u}; \boldsymbol{v})[\boldsymbol{w}] = \frac{\mathrm{d}}{\mathrm{d}\epsilon} R(\boldsymbol{u} + \epsilon \boldsymbol{w}; \boldsymbol{v}) \Big|_{\epsilon=0}.$$

The first-order expansion is

$$R(\boldsymbol{u}^k + \Delta \boldsymbol{u}^k; \boldsymbol{v}) \approx R(\boldsymbol{u}^k; \boldsymbol{v}) + \delta_u R(\boldsymbol{u}^k; \boldsymbol{v})[\Delta \boldsymbol{u}^k].$$

Setting this approximation to zero gives the linear correction problem

$$\boxed{\delta_u R(\boldsymbol{u}^k; \boldsymbol{v})[\Delta \boldsymbol{u}^k] = -R(\boldsymbol{u}^k; \boldsymbol{v}) \quad \forall \boldsymbol{v} \in \mathcal{V}_0.}$$

Update $\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + \Delta \boldsymbol{u}^k$. If the iterate satisfies the prescribed displacement, the correction is zero on Γ_{0u}. We now construct this tangent problem explicitly.

Boundary condition for the correction

Keep the prescribed displacement fixed during the nonlinear solve. Every iterate must satisfy

$$\boldsymbol{u}^k = \overline{\boldsymbol{u}} \quad \text{on } \Gamma_{0u}.$$

Substitute the update into the boundary condition for the next iterate:

$$\begin{aligned} \boldsymbol{u}^{k+1} &= \boldsymbol{u}^k + \Delta \boldsymbol{u}^k = \overline{\boldsymbol{u}}, \\ \Delta \boldsymbol{u}^k &= \overline{\boldsymbol{u}} - \boldsymbol{u}^k = \boldsymbol{0} \qquad \text{on } \Gamma_{0u}. \end{aligned}$$

Thus the correction belongs to $\mathcal{V}_0$, even when $\overline{\boldsymbol{u}} \neq \boldsymbol{0}$.
Choose the initial iterate to satisfy the prescribed displacement. Imposing $\Delta \boldsymbol{u}^k = \overline{\boldsymbol{u}}$ instead would add the prescribed displacement again at each iteration.

Differentiating the internal stress work

Hold $\boldsymbol{v}$ fixed. The prescribed loads do not depend on $\boldsymbol{u}$, so

$$\delta_u R(\boldsymbol{u}; \boldsymbol{v})[\boldsymbol{w}] = -\delta_u a_{\mathrm{hyp}}(\boldsymbol{u}; \boldsymbol{v})[\boldsymbol{w}].$$

Start with the perturbed deformation gradient:

$$\boldsymbol{F}(\boldsymbol{u} + \epsilon \boldsymbol{w}) = \boldsymbol{I} + \mathrm{Grad}\, \boldsymbol{u} + \epsilon\, \mathrm{Grad}\, \boldsymbol{w} = \boldsymbol{F}(\boldsymbol{u}) + \epsilon\, \mathrm{Grad}\, \boldsymbol{w}.$$

For a tensor $\boldsymbol{H}$, define $\delta \boldsymbol{P}(\boldsymbol{F}; \boldsymbol{H}) = \frac{\mathrm{d}}{\mathrm{d}\epsilon} \boldsymbol{P}(\boldsymbol{F} + \epsilon \boldsymbol{H})\big|_{\epsilon=0}$. Substituting $\boldsymbol{H} = \mathrm{Grad}\, \boldsymbol{w}$ and differentiating the integral gives

$$\boxed{\delta_u R(\boldsymbol{u}; \boldsymbol{v})[\boldsymbol{w}] = -\int_{\Omega_0} \delta \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u}); \mathrm{Grad}\, \boldsymbol{w}) : \mathrm{Grad}\, \boldsymbol{v}\, \mathrm{d}V_0.}$$

This step assumes enough smoothness to differentiate under the integral.

The linear variational correction problem

Insert the preceding derivative into Newton's equation:

$$\underbrace{-\int_{\Omega_0} \delta \boldsymbol{P}(\boldsymbol{F}^k; \operatorname{Grad} \Delta \boldsymbol{u}^k) : \operatorname{Grad} \boldsymbol{v} \, \mathrm{d}V_0}_{\delta_u R(\boldsymbol{u}^k; \boldsymbol{v})[\Delta \boldsymbol{u}^k]} = -R(\boldsymbol{u}^k; \boldsymbol{v}),$$

where $\boldsymbol{F}^k = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}^k$. Find $\Delta \boldsymbol{u}^k \in \mathcal{V}_0$ such that this equation holds for every $\boldsymbol{v} \in \mathcal{V}_0$. Equivalently,

$$\boxed{\delta_u R(\boldsymbol{u}^k; \boldsymbol{v})[\Delta \boldsymbol{u}^k] = -R(\boldsymbol{u}^k; \boldsymbol{v}) \quad \forall \boldsymbol{v} \in \mathcal{V}_0.}$$

For fixed $\boldsymbol{u}^k$, the left-hand side is linear in the unknown correction. Update $\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + \Delta \boldsymbol{u}^k$ and evaluate the residual again.

Identities needed for the neo-Hookean tangent

For an invertible $\boldsymbol{F}$ with $J = \det \boldsymbol{F} > 0$, take $\delta \boldsymbol{F} = \boldsymbol{H}$. Vary $\boldsymbol{F}\boldsymbol{F}^{-1} = \boldsymbol{I}$:

$$\boldsymbol{H}\boldsymbol{F}^{-1} + \boldsymbol{F}\delta(\boldsymbol{F}^{-1}) = \boldsymbol{0}, \qquad \delta(\boldsymbol{F}^{-1}) = -\boldsymbol{F}^{-1}\boldsymbol{H}\boldsymbol{F}^{-1}.$$

Transposing this identity gives

$$\boxed{\delta(\boldsymbol{F}^{-T}) = -\boldsymbol{F}^{-T}\boldsymbol{H}^{T}\boldsymbol{F}^{-T}.}$$

The determinant variation is $\delta J = J\,\mathrm{tr}(\boldsymbol{F}^{-1}\boldsymbol{H})$. The chain rule then gives

$$\boxed{\delta(\ln J) = \frac{1}{J}\delta J = \mathrm{tr}(\boldsymbol{F}^{-1}\boldsymbol{H}).}$$

All variations here correspond to the same perturbation $\delta \boldsymbol{F} = \boldsymbol{H}$.

Consistent tangent of the neo-Hookean stress

Recall

$$\boldsymbol{P}(\boldsymbol{F}) = \mu \boldsymbol{F} + (\lambda \ln J - \mu)\boldsymbol{F}^{-T}.$$

Here μ, λ are the material's Lamé constants. For $\delta\boldsymbol{F} = \boldsymbol{H}$, the product rule gives

$$\begin{aligned}\delta\boldsymbol{P}(\boldsymbol{F}; \boldsymbol{H}) = \mu \boldsymbol{H} &+ \lambda \delta(\ln J)\boldsymbol{F}^{-T} \\ &+ (\lambda \ln J - \mu)\delta(\boldsymbol{F}^{-T}).\end{aligned}$$

Substitute the two identities from the preceding slide:

$$\boxed{\begin{aligned}\delta\boldsymbol{P}(\boldsymbol{F}; \boldsymbol{H}) = \mu \boldsymbol{H} &+ \lambda \operatorname{tr}(\boldsymbol{F}^{-1}\boldsymbol{H})\boldsymbol{F}^{-T} \\ &+ (\mu - \lambda \ln J)\boldsymbol{F}^{-T}\boldsymbol{H}^{T}\boldsymbol{F}^{-T}.\end{aligned}}$$

For the variational tangent, take $\boldsymbol{F} = \boldsymbol{F}^k$ and $\boldsymbol{H} = \operatorname{Grad} \boldsymbol{w}$. *Consistent* means that this is the derivative of the stress law used in the residual.

Finite element residual and tangent matrix

The correction problem becomes a linear system for displacement coefficients.

Displacement coefficients and test functions

For N_n global nodes, use the scalar basis functions ϕ_i:

$$\boldsymbol{u}_h(\boldsymbol{X}) = \sum_{i=1}^{N_n} \phi_i(\boldsymbol{X})\boldsymbol{U}_i, \qquad \boldsymbol{U}_i = \begin{bmatrix} U_{i1} & U_{i2} & U_{i3} \end{bmatrix}^T.$$

Stack these nodal vectors into the full coefficient vector $\boldsymbol{U}$. The Cartesian unit vectors are $\boldsymbol{e}_\alpha$, $\alpha = 1, 2, 3$. A basis test direction is $\phi_i \boldsymbol{e}_\alpha$.

$$\Delta \boldsymbol{u}_h^k = \sum_{j,\beta} \phi_j \boldsymbol{e}_\beta \Delta U_{j\beta}^k, \qquad \mathrm{Grad}\, \Delta \boldsymbol{u}_h^k = \sum_{j,\beta} \Delta U_{j\beta}^k \,\mathrm{Grad}(\phi_j \boldsymbol{e}_\beta).$$

Here i, j label nodes and α, β label displacement components. The conforming FE test space $\mathcal{V}_{h,0} \subset \mathcal{V}_0$ contains these fields with zero prescribed boundary values. Only free basis directions belong to it.

Assembled residual vector

Assemble all rows before imposing prescribed coefficients. Define

$$F_{i\alpha} = \ell_{\mathrm{hyp}}(\phi_i \boldsymbol{e}_\alpha),$$
$$R_{i\alpha}(\boldsymbol{U}) = F_{i\alpha} - \int_{\Omega_0} \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u}_h)) : \mathrm{Grad}(\phi_i \boldsymbol{e}_\alpha)\, \mathrm{d}V_0.$$

Each vector entry is the variational residual evaluated on a basis direction:

$$\boxed{R_{i\alpha}(\boldsymbol{U}) = R(\boldsymbol{u}_h; \phi_i \boldsymbol{e}_\alpha).}$$

For free rows this is the variational residual tested in $\mathcal{V}_{h,0}$. On prescribed rows, the same integral defines the assembled nodal force imbalance, not a required equilibrium equation. Solve $\boldsymbol{R}_f(\boldsymbol{U}) = \boldsymbol{0}$ on the free coefficients. Prescribed coefficients are fixed by the boundary data; their residual entries are not set to zero.

Discretize first, then apply Newton's method

Differentiate the assembled residual with respect to its coefficients:

$$(K_T)_{i\alpha,j\beta}(\boldsymbol{U}) = \frac{\partial R_{i\alpha}}{\partial U_{j\beta}}.$$

Taylor expansion about $\boldsymbol{U}^k$ gives, on every free row,

$$R_{i\alpha}(\boldsymbol{U}^k + \Delta\boldsymbol{U}^k) \approx R_{i\alpha}(\boldsymbol{U}^k) + \sum_{j,\beta}(K_T)_{i\alpha,j\beta}(\boldsymbol{U}^k)\Delta U^k_{j\beta}.$$

Setting the approximation to zero gives

$$\boxed{\sum_{j,\beta}(K_T)_{i\alpha,j\beta}(\boldsymbol{U}^k)\Delta U^k_{j\beta} = -R_{i\alpha}(\boldsymbol{U}^k).}$$

$\boldsymbol{K}_T$ is the Jacobian matrix of $\boldsymbol{R}$. The loads are fixed during this iteration, so $\partial F_{i\alpha}/\partial U_{j\beta} = 0$.

Linearize first, then discretize

Start instead with $\delta_u R(\boldsymbol{u}_h^k; \boldsymbol{v}_h)[\Delta \boldsymbol{u}_h^k] = -R(\boldsymbol{u}_h^k; \boldsymbol{v}_h)$. Choose $\boldsymbol{v}_h = \phi_i \boldsymbol{e}_\alpha$ for a free row and substitute the correction expansion:

$$\sum_{j,\beta} \delta_u R(\boldsymbol{u}_h^k; \phi_i \boldsymbol{e}_\alpha)[\phi_j \boldsymbol{e}_\beta] \Delta U_{j\beta}^k = -R(\boldsymbol{u}_h^k; \phi_i \boldsymbol{e}_\alpha).$$

Since $\partial \boldsymbol{u}_h / \partial U_{j\beta} = \phi_j \boldsymbol{e}_\beta$, $\partial \boldsymbol{F} / \partial U_{j\beta} = \mathrm{Grad}(\phi_j \boldsymbol{e}_\beta)$. The stress chain rule gives

$$\boxed{(K_T)_{i\alpha j\beta} = -\int_{\Omega_0} \delta \boldsymbol{P}(\boldsymbol{F}^k; \mathrm{Grad}(\phi_j \boldsymbol{e}_\beta)) : \mathrm{Grad}(\phi_i \boldsymbol{e}_\alpha)\, \mathrm{d}V_0.}$$

The right-hand side is $-R_{i\alpha}(\boldsymbol{U}^k)$. Both routes therefore give the same correction equations when they use the same basis, quadrature, and prescribed-displacement treatment.

Element tangent entries in components

Define the fourth-order material tangent by

$$\mathbb{A}_{\alpha I \beta J}(\boldsymbol{F}) = \frac{\partial P_{\alpha I}}{\partial F_{\beta J}}, \qquad (\delta \boldsymbol{P}(\boldsymbol{F}; \boldsymbol{H}))_{\alpha I} = \sum_{\beta, J} \mathbb{A}_{\alpha I \beta J} H_{\beta J}.$$

For a local basis direction, $(\mathrm{Grad}(\phi^e_b \boldsymbol{e}_\beta))_{\gamma J} = \delta_{\gamma\beta}\, \partial \phi^e_b / \partial X_J$. Substitution gives the element tangent entry

$$\boxed{(K^e_T)_{a\alpha, b\beta} = -\int_{\mathcal{K}_e} \sum_{I,J} \frac{\partial \phi^e_a}{\partial X_I} \mathbb{A}_{\alpha I \beta J}(\boldsymbol{F}^k) \frac{\partial \phi^e_b}{\partial X_J}\, \mathrm{d}V_0.}$$

a, b are local nodes, α, β are displacement components, and I, J are reference-coordinate directions. At each iteration, evaluate $\mathbb{A}$ at the current $\boldsymbol{F}^k$ at the element integration points.

Element residuals and assembly

The element residual row is

$$R^e_{a\alpha} = F^e_{a\alpha} - \int_{\mathcal{K}_e} \sum_I P_{\alpha I}(\boldsymbol{F}^k) \frac{\partial \phi^e_a}{\partial X_I} \, \mathrm{d}V_0.$$

Here $F^e_{a\alpha}$ contains the element body-force and boundary-traction contributions. Use the existing local-to-global node map $i = I^e(a)$:

$$R_{I^e(a),\alpha} \mathrel{+}= R^e_{a\alpha},$$
$$(K_T)_{I^e(a)\alpha, I^e(b)\beta} \mathrel{+}= (K^e_T)_{a\alpha, b\beta}.$$

Begin with zero global arrays and add each element contribution. The connectivity and reference mesh stay fixed during Newton iteration. The stress and tangent values must be recomputed at the new displacement.

Free and prescribed coefficients

Partition the coefficients into free (f) and prescribed (p) entries. Since $\boldsymbol{U}_p^k = \overline{\boldsymbol{U}}_p$, the correction satisfies $\Delta \boldsymbol{U}_p^k = \boldsymbol{0}$. The free-row equations are

$$(K_T)_{ff} \Delta \boldsymbol{U}_f^k + (K_T)_{fp} \underbrace{\Delta \boldsymbol{U}_p^k}_{\boldsymbol{0}} = -\boldsymbol{R}_f(\boldsymbol{U}^k).$$

Thus solve

$$\boxed{(K_T)_{ff}(\boldsymbol{U}^k)\Delta \boldsymbol{U}_f^k = -\boldsymbol{R}_f(\boldsymbol{U}^k), \qquad \boldsymbol{U}_f^{k+1} = \boldsymbol{U}_f^k + \Delta \boldsymbol{U}_f^k.}$$

Keep $\boldsymbol{U}_p^{k+1} = \overline{\boldsymbol{U}}_p$. The internal forces depend on both free and prescribed displacements, so evaluate $\boldsymbol{R}_f$ using the *full* current vector $\boldsymbol{U}^k$.

Newton algorithm at a fixed applied load

1. Choose $\boldsymbol{U}^0$ satisfying the prescribed displacement and $J > 0$ at the integration points.
2. At $\boldsymbol{U}^k$, compute the stress, assemble $\boldsymbol{R}$ and $\boldsymbol{K}_T$, and check the free residual for convergence.
3. If it has not converged, solve

$$(K_T)_{ff}\Delta \boldsymbol{U}_f^k = -\boldsymbol{R}_f, \qquad \Delta \boldsymbol{U}_p^k = \boldsymbol{0}.$$

4. Form $\boldsymbol{U}^{k+1} = \boldsymbol{U}^k + \Delta \boldsymbol{U}^k$ and check admissibility. For the neo-Hookean law, $\ln J$ requires $J > 0$.
5. Reassemble at the new iterate. Stop on convergence or report failure at the iteration limit.

If the full update is inadmissible, it cannot simply be accepted. Step reduction and load stepping will be developed in the next lecture.

Convergence measures for the nonlinear solve

Check the free residual, since these are the equations being solved. One absolute-and-relative force criterion is

$$\boxed{\|\boldsymbol{R}_f(\boldsymbol{U}^k)\|_2 \le \varepsilon_{\mathrm{abs}} + \varepsilon_{\mathrm{rel}}\|\boldsymbol{R}_f(\boldsymbol{U}^0)\|_2.}$$

$\varepsilon_{\mathrm{abs}}$ has force units; $\varepsilon_{\mathrm{rel}}$ is dimensionless. Choose them for the force scale and desired solve accuracy. An additional displacement-change check is

$$\|\Delta \boldsymbol{U}_f^k\|_2 \le \varepsilon_u\big(U_{\mathrm{scale}} + \|\boldsymbol{U}_f^{k+1}\|_2\big),$$

where $U_{\mathrm{scale}} > 0$ is a displacement scale and ε_u is dimensionless. A small correction alone is insufficient: a stalled iteration can still have a large force imbalance. Always evaluate the updated residual.

Example: a homogeneous constrained stretch

Use the same 3D neo-Hookean material, with no body force or transverse contraction:

$$\boldsymbol{u}(\boldsymbol{X}) = \begin{bmatrix}(s-1)X_1 & 0 & 0\end{bmatrix}^T, \quad \boldsymbol{F} = \operatorname{diag}(s,1,1), \quad J = s > 0.$$

Here s is the axial stretch, and $u_1 = 0$ on $X_1 = 0$. The lateral constraints supply transverse reaction forces. On the end normal to $\boldsymbol{e}_1$, prescribe $\overline{\boldsymbol{t}}_0 = \tau \boldsymbol{e}_1$. Substitute $F_{11} = s$ into the stress law:

$$P_{11}(s) = \mu(s - s^{-1}) + \lambda \frac{\ln s}{s}.$$

The axial equilibrium equation and its Newton derivative are

$$r(s) = \tau - P_{11}(s) = 0,$$
$$r'(s) = -\mu(1 + s^{-2}) - \lambda \frac{1 - \ln s}{s^2}.$$

Example: Newton iterates for the stretch

Take $\mu = 1$ MPa, $\lambda = 2$ MPa, and $\tau = 1$ MPa. Starting from $s^0 = 1$, solve $r'(s^k)\Delta s^k = -r(s^k)$. At the first iteration,

$$r(1) = 1\ \mathrm{MPa}, \quad r'(1) = -4\ \mathrm{MPa}, \quad \Delta s^0 = 0.25, \quad s^1 = 1.25.$$

Recompute both stress and tangent after each update:

k	s^k	$r(s^k)$ (MPa)	Δs^k
0	1.000000	1.00000	0.250000
1	1.250000	0.19297	0.073251
2	1.323251	0.00913	0.003813
3	1.327064	2.15×10^{-5}	9.03×10^{-6}
4	1.327073	1.20×10^{-10}	—

The computed stress approaches the prescribed traction. The stretch remains positive throughout this iteration.

Checking a tangent by finite differences

For a free-coefficient direction $\boldsymbol{d}$ with $\boldsymbol{d}_p = \mathbf{0}$, the assembled derivative should satisfy

$$\frac{\boldsymbol{R}_f(\boldsymbol{U} + \epsilon \boldsymbol{d}) - \boldsymbol{R}_f(\boldsymbol{U})}{\epsilon} \approx (\boldsymbol{K}_T)_{ff}(\boldsymbol{U})\boldsymbol{d}_f.$$

Choose several decreasing, dimensionless ϵ values and compare the two vectors, keeping the perturbed deformation admissible. For a smooth residual, the forward-difference error initially decreases linearly with ϵ before roundoff dominates.
The scalar stretch example gives a direct check:

$$\frac{r(1.25 + \epsilon) - r(1.25)}{\epsilon} \longrightarrow r'(1.25) = -2.634376 \ \mathrm{MPa}.$$

This checks the derivative of the implemented residual. It does not by itself establish that the physical model or boundary data are correct.

Newton convergence and finite element accuracy

Let $\boldsymbol{U}^\star$ solve the nonlinear equations on the chosen mesh. If the residual has continuous second derivatives near this solution, the free tangent is nonsingular there, and the initial iterate is sufficiently close, Newton's method has local quadratic convergence:

$$\|\boldsymbol{U}_f^{k+1} - \boldsymbol{U}_f^\star\|_2 \le C\|\boldsymbol{U}_f^k - \boldsymbol{U}_f^\star\|_2^2.$$

- The Newton stopping test checks equilibrium on the chosen mesh.
- Mesh refinement changes the approximation space and addresses spatial discretization error, as in Chapters 7–8.

A small Newton residual does not demonstrate mesh convergence or selection of a physically stable equilibrium.

Summary and next lecture

Use the same residual before and after finite element discretization:

$$R(\boldsymbol{u};\boldsymbol{v}) = \ell_{\mathrm{hyp}}(\boldsymbol{v}) - a_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}),$$
$$R_{i\alpha}(\boldsymbol{U}) = R(\boldsymbol{u}_h;\phi_i\boldsymbol{e}_\alpha).$$

The corresponding correction equations are

$$\boxed{\delta_u R(\boldsymbol{u}^k;\boldsymbol{v})[\Delta\boldsymbol{u}^k] = -R(\boldsymbol{u}^k;\boldsymbol{v}), \qquad (K_T)_{ff}\Delta\boldsymbol{U}_f^k = -\boldsymbol{R}_f.}$$

- Differentiate the stress law used in the residual to obtain the tangent.
- Assemble element contributions and impose zero prescribed corrections.
- Recompute the residual after updating and distinguish solve convergence from mesh accuracy.

Next: load stepping, step reduction, and the FEniCSx/UFL implementation.

Reading

- Course book, Chapter 2, *Directional derivatives and first variations*: the derivative used with the test function held fixed.
- Course book, Chapter 4, *Finite deformation and hyperelasticity*: the stored energy, stress law, and reference variational problem.
- Course book, Chapters 5–6: coefficient vectors, assembly, prescribed coefficients, and reactions.
- Belytschko and Moran, local *Solution Methods* notes, §6.3.3–6.3.4, printed pp. 6-9–6-14: scalar and vector Newton methods, Jacobian and tangent stiffness.
- The same notes, *Convergence Criteria* and *Convergence and Robustness of Newton Iteration*, printed pp. 6-19–6-20.

For hyperelasticity, $R = \ell_{\mathrm{hyp}} - a_{\mathrm{hyp}}$ in both the variational form and the assembled residual vector.